Linna An 安琳娜, Ph.D in Chemistry
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EDUCATION

Ph.D. in Chemistry, *University of Illinois at Urbana-Champaign (UIUC), U.S.A.* ***8/2014–12/2019***

Dissertation title: The biosynthesis and discovery of lanthipeptides

Committee: Wilfred A. van der Donk (Chair), Paul J. Hergenrother, Satish K. Nair, Douglas A. Mitchell)

B.S. in Chemistry, *University of Science and Technology of China (USTC), China* ***09/2010–05/2014***

RESEARCH EXPERIENCE

Adjunct Tenure-track Assistant Faculty, *BioScience Dept, Rice University* ***08/2025–present***

Postdoctoral Scholar *for Prof.* ***David Baker****, University of Washington* ***01/2020–present***

- Project completed: Hallucination of closed repeat proteins containing central pockets
- Project completed: De novo protein design for binding and sensing small molecules
- Project completed: Protein design for enzyme specificity engineering
- Project completed: Machine learning models for small molecule-protein binding affinity ranking

Graduate Research Assistant *for Prof.* ***Wilfred A. van der Donk****, UIUC* ***08/2014–12/2019***

- Project completed: Target-guided discovery of new lanthipeptide with novel mode of action
- Project completed: Substrate-assisted enzymatic formation of lysinoalanine in duramycin
- Project completed: The mode of action of lipid II targeting two-component lantibiotics

Visiting Scholar *with Prof.* ***N. I. Martin****, Utrecht University, the Netherlands* ***10/2015–12/2015***

- Achieved isothermal calorimetry titration on lipid II-lantibiotic systems in collaboration with Dr. P. 't Hart, Dr. N. I. Martin, and Dr. E. J. Breukink.

Research Assistant *for Prof.* ***Gaolin Liang****, USTC* ***11/2011-06/2014***

- Independent Project: A new facile method for the fabrication of a 3D gold nanoparticle array with excellent surface-enhanced Raman scattering (SERS) performance
- Compound preparations and manuscript preparations for multiple molecular imaging projects

Team Member *of team USTC_China, MIT iGEM 2011* ***10/2010-11/2011***

- Project: Self-organizing Bacteria
- Individual contribution: Designing and constructing a system that is able to drive one bacterial colony into two groups based on chemotaxis. Project received Gold Award in iGEM World Jamboree at MIT

PATENT

- D. Baker, **L. An**, S. Majumder, M. Said and L. Tran, 2023, DE NOVO DESIGNED SMALL MOLECULE BINDERS VIA EXTENSIVE SHAPE COMPLIMENTARY SAMPLING, 63/610,726.

PREPRINTS (#: equal contribution, *: (co)correspondence)

1. Design of Orthogonal Far-Red, Orange and Green Fluorophore-binding Proteins for Multiplex Imaging, L. Tran, S. Sharma, S. Klein, D. Jurgens, J. Decarreau, B. Liu, Y. Wang, A. Bera, A. Kang, J. Woods, E. Joyce, D. Vafeados, N. Roullier, W. Chen, G. Lee, J. Mahamid, L. Lavis, **L. An***, D. Baker*. 2025, biorXiv, doi: https://doi.org/10.1101/2025.08.03.668343. (Preprint link)

PUBLICATIONS (#: equal contribution, *:correspondence)

2. I. Anishchenko, Y. Kipnis, I. Kalvet, G. Zhou, R. Krishna, S.J. Pellock, A. Lauko, G.R. Lee, **L. An**, J. Dauparas, F. DiMaio, & D. Baker, Modeling protein–small molecule conformational ensembles with PLACER, 2025 ***Proc. Natl. Acad. Sci. U.S.A.*** 122 (45) e2427161122. (Link)
3. G. Li#*, **L. An**#, W. Yang#, L. Yang, T. Wei, J. Shi, J. Wang, J. Doonan, K. Xie, A. R. Fernie, E. S. Lagudah, R. A. Wing, C. Gao*. Integrated biotechnological and AI innovations for crop improvement, 2025, ***Nature***, 643, 925. (Link)
4. J. Dauparas, G. Lee, R. Pecoraro, **L. An**, I. Anishchenko, C. Glasscock, D. Baker*, Atomic context-conditioned protein sequence design using LigandMPNN, 2025, ***Nature Methods***, Accepted, https://doi.org/10.1038/s41592-025-02626-1. (Link)
5. **L. An***, M. Seid, L. Tran, S. Majumder, I. Goreshnik, G. Lee, J. Dauparas, I. Anishchenko, B. Coventry, A. Bera, A. Kang, P. M. Levine, V. Alvarez, A. Pillai, C. Norn, D. Feldman, D. Zorine, D. R. Hicks, X. Li, M. G. Sanchez, D. K. Vafeados, P. J. Salveson, A. A. Vorobieva, and D. Baker*. Binding and sensing diverse small molecules using shape complementary pseudocycles, 2024, ***Science***, 385, 276. (link).
6. **L. An**#*, D. Hicks#, D. Zorine#, J. Dauparas, B. Wicky, L. Milles, A. Courbet, A. Bera, H. Nguyen, A. Kang, L. Carter, D. Baker*. Hallucination of closed repeat proteins containing central pockets, ***Nat. Struct. Mol. Biol.***, 2023, 30, 1755. (Link)
7. **L. An**#, D. Cogan#, C. Navo, G. Jiménez-Osés, S. K. Nair, and W. A. van der Donk*, Substrate-assisted enzymatic formation of lysinoalanine in duramycin, ***Nat. Chem. Biol.*** 2018, 10, 928. (Link)
8. **L. An**#* and G. R. Lee#*, De novo protein design using the blueprint builder in Rosetta, *Current Protocols in Protein Science*, 2020, 102,1, e116. (Link)
9. **L. An**, and W. A. van der Donk*, Recent progress in lanthipeptide biosynthesis, discovery,

and engineering, *Comprehensive Natural Products III*. 2019, 2, 119. (Link)

10. **L. An**, M. Rehan H. Shah Gilani, G. L. Liang*, Peptide-based nanostructures for cancer diagnosis and therapy, *Curr. Med. Chem.* 2014, 21, 2453. (Link)
11. J. Acedo, I. Bothwell, **L. An**, A. Trouth, C. Frazier, W. A. van der Donk*, O-Methyltransferase-mediated Incorporation of a β-Amino Acid in Lanthipeptides, *J. Am. Chem. Soc.* 2019, 141, 42, 16790. (Link)
12. **L. An**, "iGEM: Get & Give (& Share)", *Microbiology Today*, Web. 15 May 2018. (Link)
13. Y. Liu#, Q. Miao#, P. Zou, L. Liu, X. Wang, **L. An**, X. Zhang, X. Qian, S. Luo, and G. Liang*. Enzyme-controlled intracellular self-assembly of 18F nanoparticles for enhanced microPET imaging of tumor. *Theranostics.* 2015, 5, 1058. (Link)
14. Y. Yuan#, L. Wang#, W. Du, Z. Ding, J. Zhang, T. Han, **L. An**, H. Zhang, and G. Liang*. Intracellular self-assembly of taxol nanoparticles for overcoming multi-drug resistance. *Angew. Chem. Int. Ed*. 2015, 54, 9700. (Link)
15. Y. Yuan, J. Zhang, Q. Cao, **L. An**, G. Liang*. Intracellular disassembly of self-quenched nanoparticles turns NIR fluorescence on for sensing furin activity in cells and in tumors. *Anal. Chem*. 2015, 87, 6180. (Link)
16. Y. Yuan, S. Ge, H. Sun, X. Dong, H. Zhao, **L. An**, J. Zhang, J. Wang, B. Hu, G. Liang*. Intracellular self-assembly and disassembly of 19F nanoparticles confer respective "Off" and "On" 19F NMR/MRI signals for legumain activity detection in zebrafish. *ACS Nano*. 2015, 9, 5117-5124. (Link)
17. Y. Yuan#, H. Sun#, S. Ge#, M. Wang, H. Zhao, L. Wang, **L. An**, J. Zhang, H. Zhang, B. Hu, J. Wang, G. Liang*. Controlled intracellular self-assembly and disassembly of 19F nanoparticles for MR imaging of caspase 3/7 in zebrafish. ACS Nano. 2015, 9, 761. (Link)
18. W. Wang#, J. Qian#, A. Tang#, **L. An**, K. Zhong, G. Liang*. Using magnetic resonance imaging to study enzymatic hydrogelation. Anal. Chem. 2014, 86, 5955-5961. (Link)
19. Y. Yuan, J. Zhang, **L. An**, Q. Cao, Y Deng, G. Liang*. Oligomeric nanoparticles functionalized with NIR-emitting CdTe/CdS QDs and folate for tumor-targeted imaging. Biomaterials. 2014, 35, 7881. (Link)
20. Y. Yuan, S. Jiang, Q. Miao, J. Zhang, M. Wang, **L. An**, Q. Cao, Y. Guan, Q. Zhang, G. Liang*, Fluorescent switch for fast and selective detection of mercury (II) ions in vitro and in living cells and a simple device for its removal. Talanta. 2014, 125, 204. (Link)
21. B. You*, P. Yin, and **L. An**. (2014), Multifunctional Electroactive Heteroatom-Doped Carbon Aerogels. Small, 10: 4352-4361. (Link)
22. Y. Deng, Y. Luo, **L. An**, Y. Yue, M. Rehan H. Shah Gilani, G. Liang*. Covalently conjugating fluorescence probes to nanoparticles for signal enhancement, Chem. Lett. 2013, 42, 10, 130424. (Link)
23. Y. Yuan., X. Wang, B. Mei, D. Zhang, **L. An**, X. He, J. Jiang, G. Liang*, Labeling thiols on proteins, living cells, and tissues with enhanced emission induced by FRET, Sci. Rep. 2013, 3, 3523 (Link)

FUNDING

1. CPRIT - Recruitment of First-Time, Tenure-Track Faculty Members (RFT) award, 2025/08 - 2029/08

HONORS

2. 2024 Rising Stars in Biological Engineering, Princeton University, Omenn-Darling Bioengineering Institute, Princeton University, 09/2024
3. 2024 Rising Star in Engineering in Health, co-hosted by Boston University, Johns Hopkins University, Cornell University, Columbia University, 09/2024
4. 2024 UT Austin MBS Trailblazers of Tomorrow National Postdoctoral Symposium, 05/2024
5. New Frontiers Scholar Award, Corteva, Inc. 08/2019
6. Fuson Travel Award, University of Illinois at Urbana-Champaign, 07/2018

SELECTED Presentations

1. 2024 Workshop on Neutrons in Structural Biology, 06/2024
- Invited talk: De Novo Small Molecule Binder and Sensor Design
2. RosettaCon, 08/2023
- Presentation: De Novo Small Molecule Binder Design with Shape Complimentary Sampling
3. Biosynthesis, Biocatalysis, and New Methods in Enzymology, Gordon Research Seminar & Gordon Research Conference, 07/2023
- Presentation & Poster: De Novo Small Molecule Binder Design with Shape Complimentary Sampling
4. MRSEC seminar, invited seminar. Brandeis University 06/2023
7. Rosetta Conference, Seattle (*Invited conference, supported by RosettaCon travel grant*) 08/2022
- Poster: De novo design of small molecule binder using pseudocyclic proteins
8. Foresight's Molecular Machines Workshop, San Francisco (Invited conference) 07/2022
- Invited Talk: De novo design of small molecule binding proteins
9. New Frontiers in Natural Products Discovery conference (*Supported by New Frontiers Scholar Award),* Corteva, Inc. 08/2019
- Selected poster: A proof of concept target-guided lanthipeptide mining strategy
10. Natural Products and Bioactive Compounds Gordon Research Seminar & Gordon Research Conference *(Supported by Fuson Travel Award),* 07/2018
- Talk & poster: Substrate-assisted enzymatic formation of lysinoalanine in duramycin
11. Midwest Enzyme Chemistry Conference, Loyola University, 10/2017
- Selected talk: Substrate-assisted enzymatic formation of lysinoalanine in duramycin
11. Enzymes Coenzymes and Metabolic Pathways Gordon Research Seminar & Gordon Research Conference, 07/2017

- Poster: Mechanistic details of duramycin biosynthesis

12. Midwest Enzyme Chemistry Conference, University of Illinois at Chicago, 10/2016

- Poster: Mode of action studies for the two-peptide lantibiotic haloduracin

13. The 3rd National Chemistry Majors Scientific Activity Exchange *(Supported by the Department Travel Award)* Sichuan University, 12/2013

TEACHING & MENTORING

Mentor for rotation students, Haotian(Odin) Zhang (computer science), UW, fall/2024-present
Mentor for graduate students, Shajesh Sharma (bioengineering), UW, winter/2023-present
Mentor for rotation students, Valentina Alvarez (biochemistry), UW, summer/2023
Mentor for graduate students, Long Tran (chemical engineering), UW, winter/2022-present
Mentor for rotation student, Emma Mackey (biochemistry), UW, summer 2021
Mentor for graduate student, Chunyu Wu(biochemistry), UIUC, 09/2018-2019
Teaching Assistant for Enzymatic Reaction Mechanisms (graduate level) for Prof. J. A. Gerlt, UIUC, 01-05/2016
Teaching Assistant for General Chemistry (leading discussion and lab sessions) for Dr. K. Marville, and Dr. J. A. Martinez, UIUC, 01-05/2016, 01-06/2014, 08-12/2014
Graduate Mentor for Women Chemists Committee (WCC), UIUC, 2017

SERVICES

Women in Biology, Seattle chapter, 2022- present
Jupyter Summer Internship preparation committee, Institute for Protein Design, 2022
Women in Science, Institute for Protein Design, 2020- present
Conference Committee , Allerton Conference, UIUC, 2017
Women Chemists Committee, 2016 & 2015 Women Welcome Retreat, UIUC, 2016 & 2015
Women Chemists Committee, Stoesser Lecture, UIUC, 2016 & 2015

SKILLSETS

Computational skills: python (fluent), bash scripts, Fine tune of machine learning models, RosettaScripts (fluent), PyRosetta (fluent), machine learning-based protein design scripts (fluent), python-based data analysis (fluent).
Chemistry/Biochemistry related skills:
1. High throughput assay developments;
2. Peptide/Protein expression and purification using biological system or chemical methods;
3. Analytical methods development and use of HPLC, UV-vis, LC-HRMS (Q-TOF), GC-MS, MALDI, SEC;

4. Natural product discovery using bioinformatics & genome mining, semi-synthesis and purification, and structure elucidation using NMR and LC-MS^n;
5. Plasmid construction and gene manipulation for *Escherichia coli* and *Bacillus subtilis*.

Others:

1. Accomplished user of Illustrator, Photoshop, Autodesk 3D Max, Endnote, ChemDraw, html language;
2. Open water scuba diving;
3. Bouldering.